Sensor structure shapes the format of cognitive maps in navigation agents

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I. Background and Motivation

A PointGoal navigation (PointNav) agent is a deep-RL system that, given a goal coordinate, must walk a 3D scene to reach it. Trained agents have been observed to develop *cognitive-map-like* internal representations of space inside their recurrent (LSTM) memory — consistent with the cognitive-map hypothesis [1] and documented even in agents *without* any visual input [2], with parallels to grid- and place-cell-like codes that have been reported in recurrent navigation models [3], [4]. The agent has two routes to the spatial information its policy needs to act each step: (i) recompute it from the current visual encoder output, or (ii) maintain it across time in the recurrent memory by integrating the GPS sensor signal. These routes compete for finite hidden-state capacity. The central question of this project is whether the structure of the visual input — its encoder bandwidth and spatial layout — influences which route the trained agent ends up using, and therefore the *format* the cognitive map takes.

Our working hypothesis (the *capacity-allocation* account) is that a bottleneck at the encoder leaves the integration route as the most viable mechanism, so the LSTM tends to carry a temporally-stable, linearly-readable position code in the policy-readable layer ($\mathbf{h}_2$). A rich encoder offers a per-step visual-feature route as an alternative; under policy-gradient training, capacity tends to flow toward whichever route the agent can exploit, and the linearly-readable component of $\mathbf{h}_2$ shrinks — though, as we show, the same hidden state still encodes position non-linearly. Same task, different cognitive-map format.

This framing is an evolution of the project's original direction. The proposal and Report 1 focused on foveated vision specifically (testing whether the foveated condition shows compensatory memory). The post-retrain experiments below — particularly the foveated-logpolar control we added *after* Report 1 — pointed instead at encoder bandwidth in general as the substantive variable: foveation *per se* does not pick out a special regime. We adjusted the project scope and central question accordingly; §VI documents the shift in detail.

II. Setup and Method

We hold architecture, training, and the non-visual sensor stack constant across five conditions and vary only the visual input (Table I). All agents share a 3-layer LSTM (512-d) and the standard goal-in-start-frame / GPS / compass / close-to-goal sensor stack of [2]. We retrained all four sighted conditions from scratch under unified hyperparameters (250M frames each) to fix the inconsistency in our Report 1 runs. All achieve 96–99% task success and SPL $\in [0.85, 0.89]$ on the same evaluation set, so any cross-condition representational difference is unlikely to be explained by a skill gap.

Table I
Five conditions, ordered by encoder spatial output (size of the visual feature map per timestep). Blind uses an independently-retrained checkpoint at 340M frames.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\max} = 8$)
Uniform	4×4	256×256 full-resolution RGB

The conditions span the encoder spatial-bandwidth axis: *blind* is the no-vision lower bound replicating [2]; *coarse* is a bottleneck condition with visual input but a 1×1 encoder that collapses spatial layout per timestep; *foveated-logpolar* is an intermediate-bandwidth control we added after Report 1 as a falsification test of the encoder-bandwidth account; *foveated* and *uniform* are the two rich-encoder endpoints, distinguished by whether the periphery is degraded.

For each condition we collect 500 deterministic-policy episodes and run six probes on $\mathbf{h}_2$. A *probe* is a small regression model trained to decode some target (e.g. the agent's GPS) from $\mathbf{h}_2$ with 5-fold episode-level cross-validation; high R^2 means the target is decodable. We use both *linear* probes (Ridge α=10 — this measures what a linear policy head, the operation DD-PPO actually performs, could read) and 2-layer *MLP* probes (this measures what is encoded at all, linear-readable or not). We also run lag-k probes (decoding past position GPS_{t-k} from current $\mathbf{h}_t$), per-(unit, scene) Skaggs spatial information from the place-cell literature, cross-training probes at 5 checkpoints across training, and cross-condition CKA [5] / Procrustes shape distance / memory transplant.

III. Encoder Bandwidth Shapes the Linear Axis

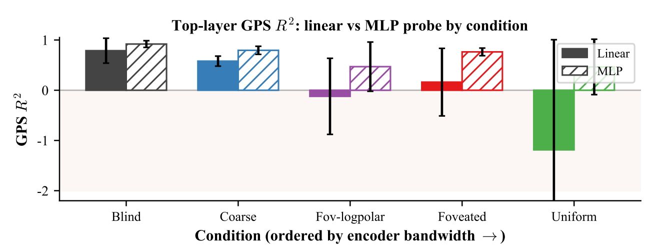

Figure 1. Top-layer GPS R^2 (mean $\pm\sigma$) vs. encoder spatial output. Solid: linear (Ridge) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

The cleanest summary of the post-retrain data is Figure 1 (solid line): linear GPS R^2 trends down smoothly as encoder spatial output grows. The trend gave us a falsification target: foveated-logpolar (a 2×2-output rich-encoder control we added *after* Report 1) was predicted to land in the rich-encoder regime, and Figure 1 is consistent with that prediction — arguing against the alternative that foveation *specifically* (rather than encoder bandwidth in general) is what shapes the cognitive map. The other obvious alternative — that the linear probe simply lacks capacity in rich-encoder hidden states — is hard to reconcile with the fact that distance-to-goal, a same-dimensional control target, decodes at $R^2 > 0.8$ in every condition.

Position information appears not to be *destroyed* by a rich encoder. A non-linear MLP probe on the *same* hidden states recovers GPS in every condition (Figure 1, dashed), and the gap from linear to MLP widens with encoder bandwidth. The combined picture is consistent with the capacity-allocation account: the bottleneck (no encoder, or 1×1) leaves the integration route as the only viable mechanism, so the LSTM tends to store position in a linearly-readable axis; a rich encoder opens a per-step visual-feature route, capacity flows toward it, and the integrated code is no longer the linearly-readable axis — though, the MLP recovery suggests, position remains encoded. Format differs; total information shifts only modestly.

IV. Allocation Emerges Across Training

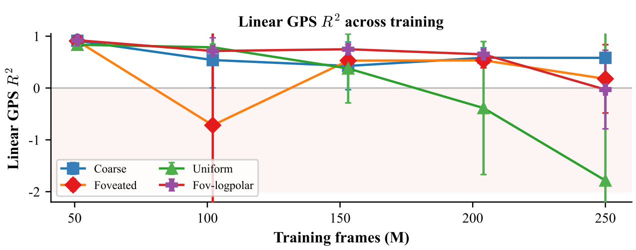

Figure 2. Linear GPS R^2 across training (5 checkpoints per condition; blind has only the 340M ckpt and is omitted). Negative-R^2 outliers clipped to -2.0.

The endpoint contrast leaves open whether bottleneck conditions ever even *learned* the visual route. Re-running the linear probe at intermediate checkpoints (Figure 2) suggests they did: all four sighted conditions begin training with a high linear top-layer GPS code, indicating that the linear axis is the *default* format the LSTM tends to acquire early on. What separates conditions is what happens *after*: bottleneck preserves the integrated code; the three rich-encoder conditions decay as the visual route consolidates, with decay rate ordered by encoder informativeness about position. To the extent the data support a mechanism, capacity allocation looks like a training-time process rather than an endpoint property; bottleneck conditions did not avoid the visual route by accident, they kept the integration route on top because no usable visual alternative emerged.

V. Bottleneck Integrates; Rich Encoder Reacts

The same format difference shows up within an episode. Decoding past position from current $\mathbf{h}_t$ (Figure 3) shows

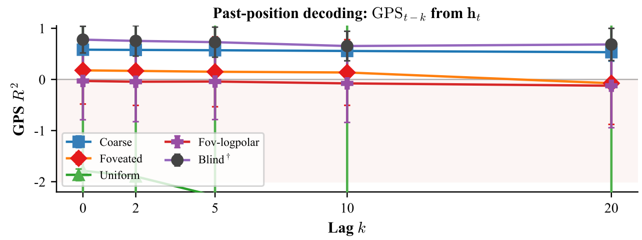

Figure 3. GPS R^2 for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Blind point: preliminary 100-ep partial NPZ.

bottleneck conditions sustain stable positive R^2 across the lag range while rich-encoder ones degrade rapidly: a smoothly-varying *integrated* code preserves past-position information; a *reactive* code keyed to current visual context does not. As a check against the obvious null — "maybe bottleneck conditions just have more place cells" — per-(unit, scene) Skaggs spatial information [2] is essentially condition-flat at $[1.16, 1.35]$ bits, well within the within-condition spread. The bandwidth-allocation signal we observe appears to live in the linear-readout axis rather than in the per-unit place-cell metric.

VI. Deviations from the Proposal

H1 reframed The proposal cast H1 as compensatory memory specifically in the foveated agent, and Report 1 (on partially-converged checkpoints with heterogeneous hyperparameters) reported foveated as the longest-memory condition. After unified retraining — which removed the apparent foveated specialness — and after adding foveated-logpolar as a falsification test, the data are more consistent with a cleaner encoder-bandwidth trend (Figure 1) under which foveated and uniform sit in the same rich-encoder regime. We now read the mechanism as encoder bandwidth rather than foveation specifically.

H3 de-scoped, methodology broadened The proposal's H3 (epistemic gaze: a learned-gaze MLP targeting memory-weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward; a bounded-σ stochastic-gaze fix is in the codebase but not pursued this iteration. To compensate for the scope reduction we imported a richer set of cognitive-neuroscience methods into the representational-analysis frame — Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, an excursion-forgetting protocol, and a leave-one-scene-out scene-invariance test. The contribution shifts from a single epistemic-gaze deliverable toward applying classical cognitive-neuroscience metrics within a computational framework for representational analysis of deep-RL agents.

VII. Open Issues and Action Items

Encountered and solved (i) Hyperparameter drift across Report 1 conditions confounded comparisons; resolved by unified retraining. (ii) Skaggs spatial information

$\sim 100\times$ inflated on raw $\mathbf{h}_2$ (near-zero global means dominate the ratio in $I = \sum p(b)(\lambda_b/\bar{\lambda}) \log_2(\lambda_b/\bar{\lambda})$); resolved by ReLU rectification ($\max(\mathbf{h}_2, 0)$) before computing per-bin firing rates. (iii) Coarse memory transplant failed at state-dict load (RGB shape mismatch); restricted to the 6 sighted-only pairs. (iv) Blind probe-collect at 50 episodes gave wide CV folds (-0.27 ± 1.41); diagnosed as sample-size noise (100 eps stabilises to $+0.79 \pm 0.25$).

Forward risks Blind 500-ep ETA overlaps the NeurIPS deadline (the 100-ep partial is a fall-back); single-seed per condition leaves seed-variance unbounded; the transplant matrix is incomplete without coarse / blind pairs, deferring the test of the proposal's H2 (cross-condition representational divergence) until an input-padding fix lands.

Bandwidth-allocation actions (A1) complete the blind 500-ep probe-collect, populate the blind row across all lenses, and tighten Figures 1 and 3; (A2) retrain coarse with a 256×256-padded input pipeline so the cross-condition memory-transplant matrix can include the bottleneck-vs-rich-encoder asymmetry test (closes the gap on H2 evidence); (A3) two-seed validation on the coarse and uniform endpoints to bound seed-variance on the magnitude axis. **Cog-neuro methodology** (A4) report the LOSO scene-invariance and excursion-forgetting results in the final write-up; (A5) place-vs-view-cell dissociation on existing rate-map data, plus an H3 reattempt with bounded-σ stochastic gaze if time permits.

VIII. Implications

If the capacity-allocation account holds up under the remaining tests, three implications follow. First, encoder bandwidth becomes a candidate *design lever* for shaping recurrent memory: deliberately restricting encoder spatial output during training may recruit integration-style memory that an unrestricted pipeline crowds out — a content-level rationale for foveation and bottleneck encoders, complementary to the usual compute-efficiency motivation [6]. Second, the encoder–memory race is in principle architecture-agnostic: vision–language models built atop rich foundation visual encoders may exhibit a similar crowd-out, plausibly contributing to the documented gap between fine-grained spatial reasoning and surface-level recognition in those systems [7]. Third, the lens we apply — linear vs non-linear probes, lag-k, place-cell signatures — is portable: it provides a way to ask, of any recurrent agent, what *kind* of cognitive map (if any) it carries, with closer ties to comparative cognitive-neuroscience methodology than is standard in deep-RL representational analysis.

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